Problem 3 Runtime Analysis

Part (a)

n	2	3	5	17
i	2	2	2	2
run time	0	1	2	3

$n=3$ run 1 time
$n=5$ run 2 times
$n=17$ run 3 times
$n=257$ run 4 times

$\log_2 \log_2 257 + 1 = 4$
$\log_2 \log_2 17 + 1 = 3$
$\log_2 \log_2 5 + 1 = 2$

$$\boxed{\log_2 \log_2 n + 1}$$

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part(b)

$$\sum_{i=1}^{n} \sum_{k=0}^{i^3-1} \theta(1)$$

$$= \sum_{i=1}^{n} \theta(i^3)$$

$$= \sum_{i=1}^{n} \theta(i^3)$$

$$= \sum_{k=1}^{\sqrt{n}} \theta(i^3)$$

$$= \sum_{k=1}^{\sqrt{n}} \theta(k^3 \sqrt{n}^3)$$

$$= \sqrt{n}^3 \sum_{k=1}^{\sqrt{n}} k^3$$

$$= \sqrt{n}^3 \, \theta(\sqrt{n}^{3+1})$$

$$= \sqrt{n}^3 \, \theta(\sqrt{n}^4) = \theta(n^{7/2})$$

K	1	2	3	4	K
i	$\sqrt{n}$	$2\sqrt{n}$	$3\sqrt{n}$	$4\sqrt{n}$	$K\sqrt{n}$

$i = k\sqrt{n}$
$k = \sqrt{n}$

$\sum_{i=1}^{n} i^3 = \frac{n^2(n+1)^2}{4}$

$n = 9$

$i = 3$ $\quad 3 \% \sqrt{9} == 0$
$i = 6$ $\quad 6 \% \sqrt{9} == 0$
$i = 9$ $\quad 9 \% \sqrt{9} == 0$

run 3 times
$\sqrt{n} = \sqrt{9} = 3$

$i = 16$ $\quad 4 \% \sqrt{16} == 0$
$8 \% \sqrt{16} == 0$
$12 \% \sqrt{}$

part (c) → 1st for loop
→ 2nd for loop

$$T(n) = \sum_{i=1}^{n} \sum_{k=1}^{n} O(1) + O\left(\sum_{m=1,2,4,8,\ldots}^{n} O(1)\right)$$

→ $m = m + m$

$$= n \cdot n + O(n \cdot \log n)$$

$$= \boxed{n^2 + O(n \log n)}$$

1st loop runs n times

2nd loop runs n times

$$n \cdot n = n^2$$

$$\sum_{m=1,2,4,\ldots}^{n} O(1) \Rightarrow n \log(n)$$

Part (d)

$$T(n) = \sum_{i=0}^{n} O(1) + \sum_{k=1}^{\log_{3/2} \frac{n}{10}} \sum_{j=1}^{10\left(\frac{3}{2}\right)^{k-1}} O(1)$$

$$= O(n) + \sum_{k=1}^{\log_{3/2} \frac{n}{10}} 10\left(\frac{3}{2}\right)^{k}$$

$$= O(n) + 10 \sum_{k=1}^{\log_{3/2} \frac{n}{10}} \left(\frac{3}{2}\right)^{k}$$

$$= O(n) + 10\Theta\left(\frac{3/2^{\log_{3/2} \frac{n}{10} + 1} - 1}{2 - 1}\right)$$

$$= O(n) + 10\Theta\left(\frac{n}{10} + 1 - 1\right)$$

$$= O(n) + \frac{10}{10} O(n)$$

$$= \boxed{O(n)}$$

$$\sum_{k=1}^{\log_{3/2} \frac{n}{10}} \Theta\left(\frac{3}{2}\right)^{k} = \left(\frac{3}{2}\right)^{\log_{3/2} \frac{n}{10}} = \Theta(n) \times \Theta(n) = \Theta(n)$$

n	1	2	3	4	5	6	7	8	9
1st for	1	2	3	4	5	6	7	8	9
if	0	0	0	0	0	0	0	0	0
2nd for	0	0	0	0	0	0	0	0	0

i	10	$\frac{3}{2}(10)$	$\left(\frac{3}{2}\right)^2(10)$		$\left(\frac{3}{2}\right)^{k-1}(10)$
k	1	2	3	...	k

$$\boxed{k \quad \left(\frac{3}{2}\right)^0(10) \quad \left(\frac{3}{2}\right)^1(10) \quad \left(\frac{3}{2}\right)^2(10) \quad \ldots \quad \left(\frac{3}{2}\right)^{k-1}(10)}$$

$$i = n - 1 = \left(\frac{3}{2}\right)^{k-1}(10)$$

$$\frac{n}{10} = \left(\frac{3}{2}\right)^{k-1} + \cancel{1} \text{ ignore}$$

$$\log_{1.5} \frac{n}{10} = k - \cancel{1} \text{ ignor}$$

$$\therefore k = \log_{1.5} \frac{n}{10}$$

$$\text{Newsize} = 10\left(\frac{3}{2}\right)^{k-1}$$

$k = 1$ $\quad i = \text{size}$

$k = 2$ $\quad i = \frac{3 \cdot \text{size}}{2}$ $\quad \frac{3}{2}$ size

$k = 3$ $\quad i = \left(\frac{3}{2}\right)^2 \text{size}$

$k = 4$ $\quad i = \left(\frac{3}{2}\right)^3 \text{size}$

$k = k$ $\quad i = \left(\frac{3}{2}\right)^{k-1} \text{size}$

$$k = \log_{1.5} \frac{n}{10}$$

$$\frac{n}{10} = \left(\frac{3}{2}\right)^{k-1}(10)$$

$$\frac{n}{10} = \left(\frac{3}{2}\right)^{k-1} \Rightarrow \log_{\frac{3}{2}} \frac{n}{10} = k - 1$$

$$k = \log_{\frac{3}{2}} \frac{n}{10} + 1$$

$$10 + \left(\frac{3}{2}\right)^1 10 + \left(\frac{3}{2}\right)^2 \times 10 + \ldots \left(\frac{3}{2}\right)^{k-1} \times 10$$

$$\Theta(\ )$$